Missing Data and Multiple Imputation

A Practical Introduction with `mice`

Monnie McGee

REU Site: Advancing Data Science in the Public Interest

Based on examples from
Flexible Imputation of Missing Data (2nd ed.)
Stef van Buuren (2018)

Sources and Acknowledgments

This lesson is adapted from:

Stef van Buuren (2018)
Flexible Imputation of Missing Data, Second Edition.

Online version:
`https://stefvanbuuren.name/fimd/`

Source code and examples adapted from:
`https://github.com/stefvanbuuren/fimdbook`

Examples have been modernized using:

- `mice`
- `naniar`
- `ggplot2`
- `VIM`

Learning Objectives

By the end of this session, you should be able to:

- Explain why missing data are a statistical problem
- Distinguish MCAR, MAR, and MNAR
- Recognize limitations of ad hoc methods
- Understand the basic idea of multiple imputation
- Use the `mice` package for a simple analysis

Why Should We Care About Missing Data?

Missing data occur everywhere:

- Surveys
- Medical studies
- Environmental monitoring
- Business data
- Social science research

Missing data can lead to:

- Smaller sample sizes
- Biased estimates
- Reduced power
- Incorrect conclusions

A Simple Example

Suppose we observe:

$$1,\ 2,\ 4$$

$$\bar{x} = 2.33$$

Now suppose we observe:

$$1,\ 2,\ \texttt{NA}$$

What should the mean be?

Quarto Demonstration

Mean calculations with missing values

Complete-Case Analysis

Also called **listwise deletion**.

Idea: remove any row with missing values.

Advantages:

- Easy
- Built into most software

Disadvantages:

- Data loss
- Potential bias
- Changing sample sizes across models

Complete-Case Analysis in R

```
lm(Ozone ~ Wind, data = airquality)

lm(Ozone ~ Wind + Solar.R, data = airquality)
```

Adding a variable can change which rows are included.

Quarto Demonstration

Listwise deletion in `airquality`

Types of Missing Data

Three missingness mechanisms:

1. MCAR
2. MAR
3. MNAR

These mechanisms help determine whether a missing-data method is reasonable.

MCAR

Missing Completely At Random

Missingness is unrelated to:

- observed values
- unobserved values

Example:

A sensor randomly fails.

Observed cases are still representative.

MAR

Missing At Random

Missingness depends on observed variables.

Example:

> *Older participants are less likely to answer a question, and age is observed.*

Multiple imputation often works well under MAR.

MNAR

Missing Not At Random

Missingness depends on the missing value itself.

Example:

People with very high income are less likely to report income.

This is the hardest case.

Think-Pair-Share

Classify each as MCAR, MAR, or MNAR:

1. A survey website crashes randomly.
2. Older respondents skip a question.
3. Students with low grades avoid reporting GPA.

Why MCAR Matters

Pairwise deletion combines statistics from different subsets.

For example:

- Mean of `Ozone` may use one set of rows
- Mean of `Solar.R` may use another set
- Covariance may use yet another set

This only works well if those subsets are representative.

That is the MCAR assumption.

Quarto Demonstration

Pairwise deletion and sample sizes

Common Ad Hoc Methods

Common quick fixes:

- Listwise deletion
- Pairwise deletion
- Mean imputation
- Regression imputation
- Last observation carried forward

They are simple, but often misleading.

Mean Imputation

Replace missing values with the mean.

$$x_{\text{missing}} = \bar{x}_{\text{observed}}$$

Advantages:

- Easy
- Fast
- Available in almost every software package

Problems:

- Reduces variance
- Distorts distributions
- Underestimates uncertainty
- Weakens relationships among variables

Quarto Demonstration

Mean Imputation

Histogram and scatterplot examples

Regression Imputation

Fit a regression model:

$$Y = \beta_0 + \beta_1 X$$

Use the fitted model to predict missing values.

Advantages:

- Uses relationships among variables
- Better than mean imputation

Problem:
Predicted values fall directly on the regression line.
Variability is lost.

Quarto Demonstration

Regression Imputation

Last Observation Carried Forward (LOCF)

Idea:
Replace a missing value by the most recent observed value.

Common in longitudinal studies.

Potential problems:

- Assumes no change occurred
- Can create artificial trends
- Often underestimates uncertainty

Quarto Demonstration

LOCF Example

Why Ad Hoc Methods Fail

Common problems:

- Loss of information
- Biased estimates
- Distorted distributions
- Incorrect standard errors
- Underestimated uncertainty

We need a method that uses available information while acknowledging uncertainty.

Multiple Imputation

Core idea:

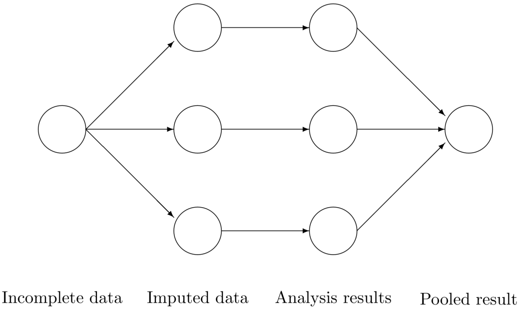
Instead of filling in one value,

fill in several plausible values.

Create:

1. Multiple completed datasets
2. Separate analyses
3. Combined results

The Multiple Imputation Workflow



What Produces the Imputed Values?

Multiple imputation is a framework.

An imputation method generates the replacement values.

Examples available in `mice`:

- Predictive mean matching (PMM)
- Bayesian regression
- Classification and Regression Trees (CART)
- Random Forest
- Logistic regression

Predictive Mean Matching (PMM)

PMM is one of the most popular methods for continuous variables.

Basic idea:

1. Predict the missing value.
2. Find observed cases with similar predictions.
3. Randomly select one observed value.

The imputed value is always an observed value from the data.

PMM Example

Suppose a student's GPA is missing.

Model prediction:

$$3.72$$

Closest observed GPAs:

$$3.70, \quad 3.80$$

PMM randomly selects one of those observed values.

Possible imputation:

$$3.70$$

or

$$3.80$$

Why PMM Is Popular

Advantages:

- Preserves realistic values
- Preserves distributions
- Avoids impossible values
- Often performs well in practice

For many continuous variables, PMM is a sensible default.

Common `mice` Methods

Method	Purpose
`pmm`	Predictive Mean Matching
`norm`	Bayesian regression
`norm.nob`	Stochastic regression
`norm.predict`	Deterministic regression
`cart`	Decision trees
`rf`	Random forest
`logreg`	Binary variables
`polyreg`	Multicategory variables
`polr`	Ordered categories

Quarto Demonstration

Using `mice`

`md.pattern()`, `mice()`, `densityplot()`, `stripplot()`

Typical mice Workflow

```
imp <- mice(
data,
method = "pmm",
m = 20,
seed = 2026
)

fit <- with(
imp,
lm(y ~ x1 + x2)
)

pool(fit)
```

What Should You Report?

When using multiple imputation:

- Number of imputations
- Imputation method
- Variables used in imputation
- Analysis model

Transparency is important.

Key Takeaways

1. Missing data are a statistical problem.
2. Missingness mechanisms matter.
3. Ad hoc methods often distort results.
4. Multiple imputation accounts for uncertainty.
5. `mice` provides practical tools for implementation.

Further Reading

Primary Reference

Stef van Buuren (2018)
Flexible Imputation of Missing Data, 2nd Edition

Book website:
`https://stefvanbuuren.name/fimd/`

R Packages

- `mice`: `https://amices.org/mice/`
- `naniar`: `https://naniar.njtierney.com/`
- `VIM`: `https://cran.r-project.org/package=VIM`

Questions?